

# AI-DISCO Benchmark Summary

Single figure summary of latency, complexity, operations breakdown, best device categories, and live pipeline timing across Jetson Orin, RTX 3050, and RTX 5090.

10/10

KPI score on every run

1.44M

Parameters

8.28

GFLOPs / inference

99.1%

Conv share of MACs

## Where Each Device Is Best

Best raw speed

RTX 5090

3.50 ms synthetic

13.79 ms live both

Best edge target

Jetson Orin

19.57 ms synthetic

27.82 ms live camera-only

50.25 ms live both

Overall ranking

So far

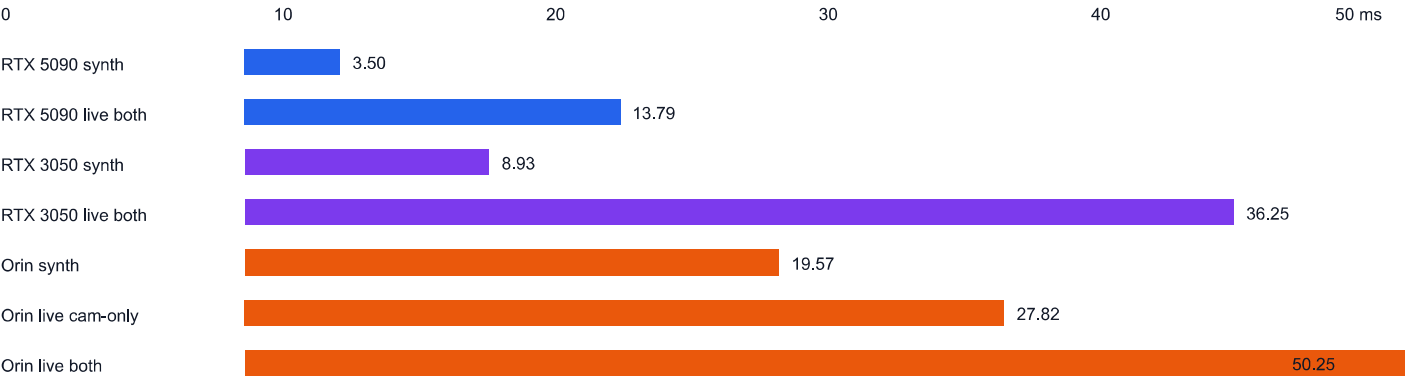
1. RTX 5090

2. RTX 3050

3. Jetson Orin

## Latency Figure

Mean latency in milliseconds. Lower is better.



## Operations Summary

Per-inference compute stays constant across all devices.

Conv 4102.96 MMACs

Temporal 12.03 MMACs

Cross-attn 24.05 MMACs

Head 0.07 MMACs

Compute share: conv dominates almost everything (~99.1%)

Model footprint: 5.51 MB weights, 5.36 MB input buffer, 1.44M params, 8.28 GFLOPs

## Live Pipeline Flow

Where time goes in real multimodal runs.

| Platform              | Radar capture (ms) | Camera preprocess (ms) | Inference (ms) | Total loop (ms) |
|-----------------------|--------------------|------------------------|----------------|-----------------|
| Jetson Orin live both | 116.6              | 15.1                   | 50.2           | 168.3           |
| RTX 3050 live both    | 163.4              | 1.4                    | 36.2           | 201.4           |
| RTX 5090 live both    | 185.6              | 0.9                    | 13.8           | 200.6           |

Main takeaway: model-forward latency is already good enough on Orin; end-to-end live throughput is mostly limited by radar acquisition cadence.

## How To Interpret The Rankings

| View               | Ranking                                   | Reason                                                                                             |
|--------------------|-------------------------------------------|----------------------------------------------------------------------------------------------------|
| Raw inference only | RTX 5090 > RTX 3050 > Jetson Orin         | Mean live-both inference latency is 13.8 ms on RTX 5090, 36.2 ms on RTX 3050, and 50.2 ms on Orin. |
| Full live loop     | Jetson Orin > RTX 5090 $\approx$ RTX 3050 | Total loop mean is 168.3 ms on Orin vs 200.6 ms on RTX 5090 and 201.4 ms on RTX 3050.              |

So Orin is not the fastest GPU, but it is currently the best full live system in this benchmark set because its radar capture stage is lower on average.

Approximate end-to-end loop rates: Orin 5.94 loops/s, RTX 5090 4.99 loops/s, RTX 3050 4.96 loops/s.

## Full Run Table

| Platform    | Run              | Mean (ms) | p50 (ms) | p95 (ms) | FPS   | KPI   |
|-------------|------------------|-----------|----------|----------|-------|-------|
| RTX 5090    | synthetic        | 3.50      | 3.37     | 4.21     | 285.9 | 10/10 |
| RTX 5090    | live both        | 13.79     | 10.16    | 23.07    | 72.5  | 10/10 |
| RTX 3050    | synthetic        | 8.93      | 8.85     | 9.32     | 112.0 | 10/10 |
| RTX 3050    | live both        | 36.25     | 20.42    | 88.26    | 27.6  | 10/10 |
| Jetson Orin | synthetic        | 19.57     | 19.15    | 22.19    | 51.1  | 10/10 |
| Jetson Orin | live camera-only | 27.82     | 24.73    | 36.59    | 35.9  | 10/10 |
| Jetson Orin | live both        | 50.25     | 56.98    | 58.74    | 19.9  | 10/10 |